

# POWER BI FOR FORENSIC ACCOUNTING & AUDIT INTELLIGENCE

A Comprehensive Hands-On Masterclass, Practical Labs & Fraud Investigation Dossier

|                            |                                                                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------|
| Course Participant:        | Tatenda Makuva (Forensic Accounting & Financial Audit Specialist)                                     |
| Dataset & Corporation:     | Apex Horizon Global Corp — Audited ERP Ledgers (FY2024 - FY2025)                                      |
| Relational Architecture:   | Star Schema: 5 Fact Tables, 6 Dimension Tables (25,830 Total Records)                                 |
| Mathematical Verification: | General Ledger Total Debits = \$77,628,218.12   Total Credits = \$77,628,218.12 (Discrepancy: \$0.00) |
| Embedded Fraud Scenarios:  | 7 Engineered Fraud & Irregularity Cases (\$1,412,297.51 Total Exposure)                               |
| Workbook Deliverables:     | 25 Step-by-Step Hands-On Tasks, Forensic DAX Guide, and Answer Verification Key                       |

**Welcome, Tatenda Makuva!** This masterclass is specifically structured for forensic accountants who want to leverage Microsoft Power BI to build automated audit tools, detect fraudulent financial patterns, and perform rigorous forensic reviews. Rather than working with superficial sample files, you will work with a fully balanced, relational enterprise ledger containing General Ledger journal entries, Accounts Payable invoices, Accounts Receivable billing, Employee T&E; reimbursements, and Bank Statements.

As you progress through the 25 hands-on lab tasks, you will uncover 7 active forensic audit anomalies, including split invoicing schemes, conflicts of interest with shell vendors, duplicate payment exploits, weekend manual journal plugs, fictitious expense claims, and unauthorized commercial discount leakage.

## Course Curriculum & Learning Roadmap:

| Module / Chapter                         | Key Concepts & Learning Objectives                                                                 | Tasks       |
|------------------------------------------|----------------------------------------------------------------------------------------------------|-------------|
| Chapter 1: Forensic Data Architecture    | Star Schema Modeling, Fact vs Dimension tables, 1-to-many single direction filtering               | Overview    |
| Chapter 2: Power Query ETL Labs          | Data Ingestion, type enforcement, whitespace trimming, off-hours flag engineering, entity matching | Tasks 1–5   |
| Chapter 3: Core Accounting DAX           | Trial Balance, Income Statement (P&L;), Balance Sheet, Gross Margin, AR Aging, YoY Variance        | Tasks 6–10  |
| Chapter 4: AP & Procurement Audit        | Split Invoicing (<\$5k limit), Shell Vendor Bank Matching, Duplicate Invoices, PO Compliance       | Tasks 11–15 |
| Chapter 5: GL & T&E; Fraud Audit         | Weekend/Midnight Manual JEs, Unapproved Plugs, Fictitious T&E; velocity, Benford's Law Analysis    | Tasks 16–20 |
| Chapter 6: Commercial Revenue Fraud      | Quarter-End Quota Gaming, Unauthorized Discounts (>20%), Row-Level Security, Capstone Dashboard    | Tasks 21–25 |
| Chapter 7: Forensic Evidence & Solutions | Case Resolution Dossier (\$1.41M Exposure), Full DAX Solution Library, Model Verification Matrix   | Solutions   |

## Chapter 1: Forensic Data Architecture & Star Schema

In forensic auditing, single-table flat files are dangerous: they blend transactional event rows with entity attributes, leading to duplicated totals, slower DAX performance, and obscured audit trails. In Power BI, we implement a **Star Schema**, placing quantitative transactions in **Fact Tables** surrounded by descriptive lookup entities in **Dimension Tables**.

### Enterprise Data Inventory (Located in the /data/ Directory):

| CSV File Name                 | Type | Rows   | Key Column(s)       | Audit Significance & Description                                                   |
|-------------------------------|------|--------|---------------------|------------------------------------------------------------------------------------|
| Dim_Date.csv                  | Dim  | 731    | Date_Key, Date      | 2024-2025 Calendar with Fiscal Periods, Quarter, and Is_Weekend flags.             |
| Dim_Chart_of_Accounts.csv     | Dim  | 32     | Account_Number      | COA with Account Class, Subclass, Normal Balance, and Sensitive flag.              |
| Dim_Cost_Centers.csv          | Dim  | 7      | Cost_Center_ID      | 7 operating departments, heads of department, and FY24/25 budgets.                 |
| Dim_Employees.csv             | Dim  | 40     | Employee_ID         | Employee master with manager hierarchy, residential address, bank account, limits. |
| Dim_Vendors.csv               | Dim  | 50     | Vendor_ID           | Approved vendor list with Tax ID, bank account, payment terms, risk rating.        |
| Dim_Customers.csv             | Dim  | 50     | Customer_ID         | Client master across 5 regions with credit limits and industry sectors.            |
| Fact_GL_Journal_Entries.csv   | Fact | 18,579 | Journal_ID, Line_ID | Balanced double-entry journal ledger (\$77.63M debits/credits).                    |
| Fact_Vendor_Invoices_AP.csv   | Fact | 1,734  | Invoice_ID          | AP invoices (\$8.48M spend), PO match status, approver ID, payment terms.          |
| Fact_Sales_Invoices_AR.csv    | Fact | 2,881  | Sales_Invoice_ID    | Customer invoices (\$32.07M gross/net sales), discount amounts, rep IDs.           |
| Fact_Employee_Expenses_TE.csv | Fact | 1,833  | Expense_ID          | T&E; reimbursement claims, merchants, receipt attachment flags.                    |
| Fact_Bank_Statements.csv      | Fact | 72     | Bank_Trans_ID       | Monthly treasury bank statement feed for cash account #1010 reconciliation.        |

### Model View Relationship Map (All 1-to-Many Single-Direction Filtering):

| Primary Dimension (1)                  | Target Fact Table (*)                    | Key Join Column            | Cardinality & Direction   |
|----------------------------------------|------------------------------------------|----------------------------|---------------------------|
| Dim_Date [Date]                        | Fact_GL_Journal_Entries [Posting_Date]   | Date → Posting_Date        | 1:*   Single (Dim → Fact) |
| Dim_Date [Date]                        | Fact_Vendor_Invoices_AP [Invoice_Date]   | Date → Invoice_Date        | 1:*   Single (Dim → Fact) |
| Dim_Date [Date]                        | Fact_Sales_Invoices_AR [Invoice_Date]    | Date → Invoice_Date        | 1:*   Single (Dim → Fact) |
| Dim_Date [Date]                        | Fact_Employee_Expenses_TE [Expense_Date] | Date → Expense_Date        | 1:*   Single (Dim → Fact) |
| Dim_Chart_of_Accounts [Account_Number] | Fact_GL_Journal_Entries [Account_Number] | Account_Number             | 1:*   Single (Dim → Fact) |
| Dim_Cost_Centers [Cost_Center_ID]      | Fact_GL_Journal_Entries [Cost_Center_ID] | Cost_Center_ID             | 1:*   Single (Dim → Fact) |
| Dim_Cost_Centers [Cost_Center_ID]      | Fact_Vendor_Invoices_AP [Cost_Center_ID] | Cost_Center_ID             | 1:*   Single (Dim → Fact) |
| Dim_Vendors [Vendor_ID]                | Fact_Vendor_Invoices_AP [Vendor_ID]      | Vendor_ID                  | 1:*   Single (Dim → Fact) |
| Dim_Customers [Customer_ID]            | Fact_Sales_Invoices_AR [Customer_ID]     | Customer_ID                | 1:*   Single (Dim → Fact) |
| Dim_Employees [Employee_ID]            | Fact_Employee_Expenses_TE [Employee_ID]  | Employee_ID                | 1:*   Single (Dim → Fact) |
| Dim_Employees [Employee_ID]            | Fact_Sales_Invoices_AR [Sales_Rep_ID]    | Employee_ID → Sales_Rep_ID | 1:*   Single (Dim → Fact) |

#### CRITICAL MODELING STANDARD

**Forensic Golden Rule:** Never enable Bi-Directional cross-filtering across multiple Fact tables. Bi-directional filtering creates ambiguity in filter propagation, degrades performance, and can generate artificial duplicate aggregations. Keep filter flows moving in one direction: from Dimension tables outward to Fact tables.

## Chapter 2: Power Query (ETL) & Data Cleansing Labs

In forensic investigations, raw source data cannot be trusted blindly. Fraudsters often manipulate whitespace, casing, and time stamps to evade automated controls. Perform these 5 essential ETL tasks in **Power Query Editor**:

### Task 1.1: Automated Folder Ingestion & Data Type Verification

1. In Power BI Desktop, navigate to **Home > Get Data > Folder** and point to the /data/ folder.
  2. Ensure all 11 tables are loaded. Verify that Date columns are typed as **Date**, dollar amounts as **Fixed Decimal Number (\$)**, and Account Numbers (e.g., 1010, 2010) as **Text**.
- Why is typing Account Numbers as Text critical?* (Prevents Power BI from accidentally summing account codes as numbers in visuals).

### Task 1.2: Mark Official Date Table & Build Chronological Hierarchies

1. In Data View, select Dim\_Date. In the ribbon, select **Table Tools > Mark as Date Table** with identifier Date.
2. Select Month\_Name, navigate to **Column Tools > Sort by Column**, and select Month\_Num. This ensures calendar charts display months in proper chronological sequence rather than alphabetical order.
3. Create a 4-level Date Hierarchy: Fiscal\_Year > Fiscal\_Quarter > Month\_Name > Date.

### Task 1.3: Forensic Whitespace Trimming (Exposing Duplicate Exploit)

1. In Power Query Editor, open Fact\_Vendor\_Invoices\_AP.
  2. Right-click the Invoice\_ID column and select **Transform > Trim** and **Transform > Clean**.
- Forensic Investigation Finding:** Notice invoice AP-INV-92011 contained a sneaky trailing space! Without trimming, traditional VLOOKUP or relational joins treat it as a distinct invoice, allowing a duplicate payment of **\$19,750.50** to be processed unchecked.

### Task 1.4: Feature Engineering — Off-Hours & Weekend Posting Flags

1. In Fact\_GL\_Journal\_Entries, split the Posting\_Timestamp column by space delimiter into Posting\_Date\_Only and Posting\_Time.
  2. Navigate to **Add Column > Custom Column** named Is\_Off\_Hours\_Posting with the M formula:  
 if (Time.Hour([Posting\_Time]) < 7 or Time.Hour([Posting\_Time]) >= 19) or [Is\_Weekend] = 1 then 1 else 0.
- This flags all journal adjustments made late at night or over weekends for immediate forensic review.

### Task 1.5: Cross-Entity Matching Merge (Employee vs. Vendor Master)

1. In Power Query, select **Home > Merge Queries as New**.
  2. Select Dim\_Employees on top and Dim\_Vendors on the bottom, joining on Bank\_Account\_Number with an **Inner Join**.
- Forensic Discovery Check:** Identify the exact employee and vendor match! (Employee **EMP-118 David Vance** and Vendor **VND-1038 Apex Global Consulting** share bank account ACCT-US-908812 and residential address 742 Evergreen Terrace).

#### AUDIT TRAIL BEST PRACTICE

**Power Query Audit Best Practice:** Always rename Applied Steps to reflect your investigative hypothesis (e.g., Trimmed\_AP\_Invoice\_Keys, Flagged\_Off\_Hours\_Timestamps, Merged\_Employee\_Vendor\_Bank\_Match). This ensures that court evidence and audit workpapers maintain a complete, reproducible chain of custody.

## Chapter 3: Core Accounting & Financial DAX Formulas

Create a dedicated blank measures table named `_Forensic_Measures` in Power BI Desktop (via **Enter Data**). Implement the following financial and double-entry accounting measures:

### 1. Double-Entry Balance Validator Measures

```
Total Debits = SUM(Fact_GL_Journal_Entries[Debit])
Total Credits = SUM(Fact_GL_Journal_Entries[Credit])
GL Net Discrepancy Check = [Total Debits] - [Total Credits]
GL Balance Status = IF(ROUND([GL Net Discrepancy Check], 2) = 0, "■ BALANCED TO THE CENT", "■ IMBALANCE: " & FORMAT([GL Net Discrepancy Check], "$#,##0.00"))
```

### 2. Trial Balance Net Activity & P&L; Core Metrics

```
GL Net Activity = SUMX(Dim_Chart_of_Accounts, IF(Dim_Chart_of_Accounts[Normal_Balance] = "Debit", [Total Debits] - [Total Credits], [Total Credits] - [Total Debits]))
Gross Commercial Revenue = SUM(Fact_Sales_Invoices_AR[Gross_Sales_Amount])
Total Sales Discounts = SUM(Fact_Sales_Invoices_AR[Discount_Amount])
Net Sales Revenue = SUM(Fact_Sales_Invoices_AR[Net_Sales_Amount])
Total Cost of Goods Sold = SUM(Fact_Sales_Invoices_AR[Cost_of_Sales])
Gross Profit = [Net Sales Revenue] - [Total Cost of Goods Sold]
Gross Margin % = DIVIDE([Gross Profit], [Net Sales Revenue], 0)
```

### 3. Operating Expenses & EBITDA Operating Income

```
Total Operating Expenses = CALCULATE([Total Debits] - [Total Credits], Dim_Chart_of_Accounts[Account_Subclass] = "Operating Expenses")
EBITDA Operating Income = [Gross Profit] - [Total Operating Expenses]
```

## Module 2 Hands-On Tasks (Financial Performance Dashboards):

### Task 6: Interactive Double-Entry Trial Balance Matrix

Build a Matrix visual with `Account_Class > Account_Subclass > Account_Name` on Rows. Add `[Total Debits]`, `[Total Credits]`, and `[GL Net Activity]` to Values.

**Verification Checkpoint:** Confirm that Total Debits equal exactly **\$77,628,218.12** and Total Credits equal **\$77,628,218.12** (Discrepancy = \$0.00).

### Task 7: Dynamic Income Statement (P&L;) with YoY Growth

Create a visual showing Net Sales Revenue (\$32.07M), COGS (\$16.61M), Gross Profit (\$15.46M, 48.2%), and Operating Expenses. Add a DAX measure for Net Revenue YoY % comparing FY2025 to FY2024.

### Task 8: Cost Center Budget vs. Actual Variance Gauge

Create a Gauge visual comparing department actual spend against `Budget_FY24` and `Budget_FY25` from `Dim_Cost_Centers`. Which departments exceeded budget?

### Task 9: Accounts Receivable Aging & Bad Debt Exposure

Build a 100% Stacked Bar chart breaking down outstanding AR invoices across aging buckets (Current, 1-30, 31-60, 61-90, 90+ days). Total bad debt written off = \$42,100+.

### Task 10: Vendor Spend Pareto Curve (80/20 Rule)

Build a Line & Clustered Column Chart calculating cumulative spend % across vendors to identify the top 20% critical vendor concentration.

## Chapter 4: Accounts Payable & Procurement Forensic Audit

Procurement fraud accounts for over 30% of enterprise occupational fraud losses (ACFE Report to the Nations). In this module, you will write specialized forensic DAX measures to detect split purchases, ghost shell vendors, and duplicate billings.

### Forensic DAX 1: Split Invoice Structuring Below Approval Limit (\$5,000 Threshold)

```
-- Detects invoices structured between $4,850 and $4,999 to bypass departmental VP authorization limit
Count Split Invoices (<$5k Limit) = CALCULATE(COUNTROWS(Fact_Vendor_Invoices_AP), Fact_Vendor_Invoices_AP[Invoice_Amount] >= 4850, Fact_Vendor_Invoices_AP[Invoice_Amount] < 5000)
Split Invoices Total Spend = CALCULATE(SUM(Fact_Vendor_Invoices_AP[Invoice_Amount]), Fact_Vendor_Invoices_AP[Invoice_Amount] >= 4850, Fact_Vendor_Invoices_AP[Invoice_Amount] < 5000)
```

### Forensic DAX 2: Potential Duplicate Payments Measure

```
Duplicate Invoiced Spend =
SUMX(
    SUMMARIZE(Fact_Vendor_Invoices_AP, Fact_Vendor_Invoices_AP[Vendor_ID], Fact_Vendor_Invoices_AP[Invoice_Amount],
        "InvoiceCount", COUNT(Fact_Vendor_Invoices_AP[Invoice_ID]), "SumAmount", SUM(Fact_Vendor_Invoices_AP[Invoice_Amount])),
    IF([InvoiceCount] > 1, [SumAmount] - ([SumAmount] / [InvoiceCount]), 0)
)
```

## Module 3 Practical Tasks (AP & Procurement Audit):

### Task 11: Exposing the Split Invoicing Scheme (CloudSphere Solutions VND-1042)

Filter Fact\_Vendor\_Invoices\_AP for vendor CloudSphere Solutions LLC (VND-1042).

**Forensic Investigation Findings: 48 invoices** totaling **\$236,412.88** approved by **EMP-103 Marcus Brody** (VP Commercial Sales) structured between \$4,853.11 and \$4,992.47 to bypass Brody's \$5,000 limit with 0 PO matching.

### Task 12: Exposing Conflict of Interest & Shell Vendor (Apex Global Consulting VND-1038)

In your data model, examine Apex Global Consulting Ltd (VND-1038).

**Forensic Investigation Findings: 13 invoices** totaling **\$266,129.04** paid with 0 PO. Created and approved by **EMP-118 David Vance** (Senior Procurement Officer), sharing identical bank account ACCT-US-908812 and address 742 Evergreen Terrace!

### Task 13: Detecting Duplicate Invoice Payments (Summit Logistics VND-1015)

Create a Table visual filtered on Vendor\_ID = 'VND-1015' displaying Invoice\_ID, Invoice\_Date, Invoice\_Amount, and Payment\_Method.

**Forensic Discovery: 4 duplicate invoice pairs** paid twice (\$18,450.00, \$24,800.00, \$19,750.50, \$31,200.00) totaling **\$94,200.50** in excess paid.

### Task 14: Non-PO Emergency Disbursement Audit

Calculate % of AP spend with Is\_PO\_Matched = 'No'. Supply Chain (CC-500) and Sales (CC-300) account for the vast majority of non-PO disbursements.

### Task 15: Dynamic Vendor Fraud Risk Scorecard

Create a calculated measure assigning a 1-100 Risk Score based on High Risk Rating (+30), Split Invoicing (+30), Non-PO Payments (+20), and Duplicate History (+20).

## Chapter 5: General Ledger & Travel Expense Fraud Audit

Manual journal entries and employee expense claims are primary vehicles used to conceal fraudulent disbursements and manipulate financial statements. Use Power BI to detect unauthorized weekend adjustments and fictitious expense velocity spikes.

### Forensic DAX 3: Weekend & Off-Hours Manual Journal Adjustments

```
-- Quantifies manual journal adjustments posted on weekends or between 20:00 and 06:00
Weekend Manual JEs Total = CALCULATE(SUM(Fact_GL_Journal_Entries[Debit]), Fact_GL_Journal_Entries[Is_Manual] = 1, Dim_Date[Is_Weekend] = 1)
Unapproved Suspicious Adjustments = CALCULATE(SUM(Fact_GL_Journal_Entries[Debit]), Fact_GL_Journal_Entries[Is_Manual] = 1, ISBLANK(Fact_GL_Journal_Entries[Approved_By]) || Fact_GL_Journal_Entries[Approved_By] = "")
```

### Forensic DAX 4: Missing Receipt Rate in Travel & Entertainment (T&E;)

```
Missing Receipt Claims Amount = CALCULATE(SUM(Fact_Employee_Expenses_TE[Claim_Amount]), Fact_Employee_Expenses_TE[Receipt_Attached] = "No")
Missing Receipt % = DIVIDE([Missing Receipt Claims Amount], SUM(Fact_Employee_Expenses_TE[Claim_Amount]), 0)
```

## Module 4 Practical Tasks (GL & T&E; Investigation):

### Task 16: Weekend & Off-Hours Manual Journal Entry Investigation

Filter Fact\_GL\_Journal\_Entries for Is\_Manual = 1 and Dim\_Date [Is\_Weekend] = 1.

**Forensic Findings:** 8 unique manual entries debited exactly **\$533,000.00**, posted by **EMP-105 Elena Rostova** on Saturday nights (23:35-23:55) with **blank approvers**!

### Task 17: Deep-Dive Audit of Account 9100 (Miscellaneous Expense)

Drill down into Account 9100 (Miscellaneous & Sundry Expense). Notice round-dollar debit spikes (\$45k, \$62k, \$78k, \$49k, \$67k, \$95k) offsetting cash account 1010.

### Task 18: Exposing Fictitious T&E; Claims Velocity Spike (Julian Thorne EMP-129)

In Fact\_Employee\_Expenses\_TE, create a Matrix with Employee\_ID on Rows, and Claim Count, Total Amount, and Missing Receipt Count.

**Forensic Findings:** Julian Thorne submitted **208 claims totaling \$30,410.19** on consecutive weekends with **100% missing receipts** structured between \$142.50 and \$149.95 to evade the mandatory \$150 receipt rule!

### Task 19: Benford's Law First-Digit Distribution Visual

Create a Bar Chart displaying the frequency distribution of the first digit (1-9) of GL expense debits compared to Benford's Law theoretical benchmarks (Digit 1 = 30.1%, Digit 2 = 17.6%). Account 9100 exhibits massive artificial clustering in digits 4, 6, and 7.

### Task 20: GL to Bank Treasury Cash Reconciliation

Build a reconciliation matrix comparing Fact\_Bank\_Statements monthly net deposits/withdrawals against General Ledger cash account 1010. Verify that unreconciled timing differences are flagged automatically.

## Chapter 6: Commercial Revenue Fraud & Capstone Dashboard

Commercial revenue fraud often involves sales reps offering unauthorized steep discounts to reach commission quotas or collude with favored buyers. In this module, you will quantify commercial margin leakage, establish Row-Level Security, and build the Executive Capstone Dashboard.

### Forensic DAX 5: Unauthorized High-Discount Margin Leakage

```
-- Quantifies total revenue lost to sales discounts exceeding the 15% corporate maximum policy
Unauthorized High Discounts Amount = CALCULATE(SUM(Fact_Sales_Invoices_AR[Discount_Amount]), Fact_Sales_Invoices_AR[Discount_Percentage] > 0.15)
High Discount Invoices Count = CALCULATE(COUNTROWS(Fact_Sales_Invoices_AR), Fact_Sales_Invoices_AR[Discount_Percentage] > 0.15)
```

### Module 5 Practical Tasks & Capstone Build:

#### Task 21: Sales Rep Discount Policy Deviation (Rachel Zane EMP-133)

Build a Scatter Plot with Sales Reps on Details, Gross Sales on X-Axis, and Average Discount % on Y-Axis.

**Forensic Findings:** While company sales reps average a **3.1%** discount rate, **Rachel Zane (EMP-133)** averages **7.4%** overall and has **55 invoices** with extreme discounts exceeding 20%.

#### Task 22: Correlating Quarter-End Timing with Extreme Discounts

Plot Rachel Zane's discount percentages over time on a Line Chart with Dim\_Date [Date].

Notice the unmistakable surge in discounts (30% to 42%) occurring exclusively in the final 10 days of Q1 (March), Q2 (June), Q3 (September), and Q4 (December) — classic quota manipulation!

#### Task 23: Commercial Margin Leakage Dollar Impact

Calculate the total net revenue lost to Rachel Zane's unauthorized discount spikes. Total high-discount loss = **\$252,144.90**.

#### Task 24: Enterprise Row-Level Security (RLS) Configuration

In Power BI Desktop ribbon, navigate to **Modeling > Manage Roles**. Create two roles:

1. **Department\_Manager**: DAX Filter on Dim\_Cost\_Centers: [Cost\_Center\_ID] = USERPRINCIPALNAME().
2. **Forensic\_Auditor**: No filter (full unrestricted visibility across all 11 tables).

Test the roles using **View as Roles** to ensure security rules enforce strict segregation.

#### Task 25: Master Forensic Intelligence Capstone Dashboard

Construct a publication-grade, 4-page Power BI Dashboard report:

- **Page 1: Executive Audit Summary**: KPI Cards for Total Debits/Credits, Net Sales, Total AP, and Discovered Fraud Exposure (\$1.41M).
- **Page 2: Accounts Payable Fraud Engine**: Split invoice scatter plot, Shell vendor bank match table, Duplicate invoice tracker.
- **Page 3: General Ledger Journal Anomaly Matrix**: Weekend/Off-hours heatmap, Account 9100 drill-down, Benford's Law bar chart.
- **Page 4: T&E; & Commercial Leakage**: Sales rep discount outliers, Julian Thorne weekend expense velocity tracker.

## Chapter 7: Forensic Audit Evidence Summary & Case Dossier

The following forensic investigation dossier summarizes the 7 fraud and irregularity schemes uncovered across Apex Horizon Global Corp. Use this evidence table to verify your analytical findings and prepare executive audit briefs.

| Scheme & Case Name                                | Key Perpetrator(s)                  | Financial Exposure               | Audit Evidence Trail & Modus Operandi                                                                                                                                                                             |
|---------------------------------------------------|-------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Case 1: Split Invoicing Structuring               | Marcus Brody (EMP-103, VP Sales)    | \$236,412.88 (48 Invoices)       | Invoices from CloudSphere Solutions (VND-1042) billed at \$4,850-\$4,995 in rapid 2-day clusters to bypass Brody's \$5,000 approval limit. Zero formal PO matching.                                               |
| Case 2: Ghost Shell Vendor & Conflict of Interest | David Vance (EMP-118, Procurement)  | \$266,129.04 (13 Invoices)       | David Vance registered shell company Apex Global Consulting (VND-1038) using his personal residence (742 Evergreen Terr) and bank account (ACCT-US-908812). Billed vague advisory retainers with immediate terms. |
| Case 3: Duplicate Invoices & Overpayments         | Summit Logistics (VND-1015)         | \$94,200.50 (4 Duplicate Pairs)  | Duplicate freight invoices submitted with slight variations (trailing spaces, 'DUP' and 'A' suffixes) paid twice via EFT and Check runs.                                                                          |
| Case 4: Off-Hours Manual Journal Plugs            | Elena Rostova (EMP-105, Senior GL)  | \$533,000.00 (8 Manual JEs)      | Manual journal entries posted on Saturday nights (23:35-23:55) debiting Account 9100 (Miscellaneous) and crediting Operating Cash (1010). No manager approval attached.                                           |
| Case 5: Fictitious Weekend T&E; Claims            | Julian Thorne (EMP-129, Sales Rep)  | \$30,410.19 (208 Claims)         | 208 consecutive weekend dining/lounge claims submitted at \$142-\$149.95 to stay just below the mandatory \$150 receipt audit rule. 100% missing receipts.                                                        |
| Case 6: Unauthorized High Discounts               | Rachel Zane (EMP-133, Key Accounts) | \$252,144.90 (55 Invoices > 20%) | Applied unauthorized discounts of 30% to 42% at quarter-ends (March, June, Sept, Dec) to artificially inflate volume and achieve performance bonus thresholds.                                                    |
| TOTAL FINANCIAL EXPOSURE IDENTIFIED               | Multiple Internal Actors            | \$1,412,297.51                   | Comprehensive internal control failure requiring immediate CFO notification, vendor freeze, and internal control segregation of duties (SoD) redesign.                                                            |

### Recommended Internal Control & Governance Remediation Actions:

- Automated SoD & Dual Approval Limits:** Implement hard system locks preventing cost center managers from approving invoices within 10% of their authorization threshold without secondary VP sign-off.
- ERP Master Data Matching:** Establish automated nightly cross-join checks between the Employee bank account / residential address database and Vendor master records.
- Strict 3-Way Matching:** Mandate 3-way matching (PO, Receiving Report, Invoice) for all service and consulting invoices exceeding \$2,500.
- GL Posting Window Locks:** Restrict manual General Ledger posting privileges to weekdays between 07:00 and 19:00, requiring explicit dual-key CFO approval for off-hours adjustments.
- Zero-Receipt T&E; Rejection:** Enforce automatic rejection for any expense claim lacking an attached itemized digital receipt, regardless of dollar threshold.

## Chapter 8: DAX Solutions & Model Verification Key

Use this complete solution dictionary and verification matrix to validate your Power BI calculations against the mathematical ground truth:

| Metric / Measure Name              | Expected Value / Ground Truth | Verification Visual & Filter Context                        |
|------------------------------------|-------------------------------|-------------------------------------------------------------|
| GL Total Debits                    | \$77,628,218.12               | Card visual: SUM(Fact_GL_Journal_Entries[Debit])            |
| GL Total Credits                   | \$77,628,218.12               | Card visual: SUM(Fact_GL_Journal_Entries[Credit])           |
| GL Net Discrepancy                 | \$0.00 (Balanced)             | Card visual: [Total Debits] - [Total Credits]               |
| Total Net Sales Revenue            | \$32,066,469.23               | Card visual: SUM(Fact_Sales_Invoices_AR[Net_Sales_Amount])  |
| Total Cost of Goods Sold           | \$16,605,945.19               | Card visual: SUM(Fact_Sales_Invoices_AR[Cost_of_Sales])     |
| Gross Profit (\$ and %)            | \$15,460,524.04 (48.2%)       | Card visual: [Net Sales Revenue] - [Total COGS]             |
| Total AP Invoiced Spend            | \$8,476,029.18 (1,734 invs)   | Matrix visual: SUM(Fact_Vendor_Invoices_AP[Invoice_Amount]) |
| CloudSphere Split Invoices         | \$236,412.88 (48 invs)        | Filter AP on Vendor_ID = 'VND-1042'                         |
| Apex Global Shell Vendor           | \$266,129.04 (13 invs)        | Filter AP on Vendor_ID = 'VND-1038'                         |
| Summit Logistics Excess Duplicates | \$94,200.50 (4 pairs)         | Filter AP on Vendor_ID = 'VND-1015' duplicate amounts       |
| Off-Hours Manual JEs               | \$533,000.00 (8 journals)     | Filter GL on Is_Manual = 1 & Is_Weekend = 1                 |
| Julian Thorne Fake T&E;            | \$30,410.19 (208 claims)      | Filter TE on Employee_ID = 'EMP-129'                        |
| Rachel Zane Excess Discounts       | \$252,144.90 (55 invs > 20%)  | Filter AR on Sales_Rep_ID = 'EMP-133' & disc > 20%          |
| Total Discovered Fraud Exposure    | \$1,412,297.51                | Sum of all 6 forensic exposure measures                     |

### PROFESSIONAL ATTAINMENT

**Congratulations on Completing the Masterclass, Tatenda!**

By completing this workbook and building the complete Power BI Forensic Audit Model, you have mastered the transition from traditional audit procedures to modern, automated forensic data analytics. You now possess the skills to model complex financial schemas, write advanced forensic DAX expressions, and build interactive intelligence dashboards that safeguard enterprise assets.